(Please write your Enrolment No. immediately) Enrolment No.....

**SUPPLEMENTARY MID TERM EXAMINATION**
**B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)**
**Fourth Semester, March 2025**

**Paper Code: BS-202** **Subject: Probability, Statistics and Linear Programming**
**Time: $1\frac{1}{2}$ Hrs.** **Max. Marks: 30**

**Note: Attempt O.No.1, which is compulsory, and any two more questions from remaining.**

| Q. No. | Question | Max. Marks | CO(s) |
|---|---|---|---|
| 1(a) | A problem in mathematics is given to three students $A$, $B$, and $C$ whose chance of solving it are 0.5, 0.3 and 0.25 respectively. What is the probability that the problem will be solved? | 2.5 | CO1 |
| 1(b) | In a large corporate computer network, user log-ons to the system can be modelled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes? | 2.5 | CO1 |
| 1(c) | State the Central Limit Theorem. Also, find the standard deviation of the sampling distribution if the sample size is 36 and the population standard deviation is 12? | 2.5 | CO2 |
| 1(d) | A random variable $X$ has the probability mass function $p(x) = \frac{2}{3^x}, x = 1,2,3, \ldots .$ Find its moment generating function about origin. | 2.5 | CO2 |
| 2(a) | Determine the value of $k$ and $E(X^2)$ if the probability density function of a random variable $X$ is given by $f(x) = \begin{cases} 2k & ; 0 \le x < 2 \\ -kx + 4k & ; 2 \le x < 4 \\ 0 & ; \text{otherwise} \end{cases}$ . | 5 | CO1 |
| 2(b) | In a normal distribution, 8% of the items are below 40, and 85% are below 70. Find the mean and standard deviation. Given that: $P(Z < 1.41) = 0.92$ and $P(Z < 1.04) = 0.85$ | 5 | CO1 |
| 3(a) | Find the value of $c$ such that $f(x, y) = cxy$ for $0 < x < 3$ and $0 < y < x$ is a joint probability density function of two random variables $X$ and $Y$. Also, find Conditional probability distribution of $X$ given $Y = 1$. | 5 | CO2 |
| 3(b) | Find the covariance for the following joint probability distribution: | 5 | CO2 |
| 4(a) | Consider the completion time of a large commercial development. The proportion of the maximum allowed time to complete a task is modelled as a beta random variable with $\alpha = 2.5$ and $\beta = 1$. What is the probability that the proportion of the maximum time exceeds 0.7? Also, find the mean of the random variable. | 5 | CO1 |
| 4(b) | Suppose that $X$ is a random variable with mean $\mu$ and variance $\sigma^2$. Let $X_1, X_2, \ldots . X_n$ be a random sample of size $n$ from the population represented by $X$. Show that sample variance $S^2$ is an unbiased estimator of population variance $\sigma^2$ . | 5 | CO2 |

| $x$ | 1 | 1 | 2 | 4 |
|---|---|---|---|---|
| $y$ | 3 | 4 | 5 | 6 |
| f($x$,$y$) | 1/8 | 1/4 | 1/2 | 1/8 |